

Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

Ch 3 · Analysis of LTI Systems in the Frequency Domain

Numerical Problems & Solutions

35 of the 56 questions in this chapter are the same question: plot the poles and zeros, then sketch the magnitude response. All of them are below. They split into two shapes, according to whether the paper gives you a difference equation or the pole and zero locations themselves, and the second half of the work is identical in both.

The one procedure, used in §1 and §2

- 1. Get $H(z)$.** From an LCCDE, Z-transform both sides and divide. Multiply top and bottom by z^N so both are polynomials in z , not in z^{-1} .
- 2. Factor.** Numerator roots are zeros, denominator roots are poles. These are almost always quadratics, so the quadratic formula finishes it, and complex roots always arrive in conjugate pairs.
- 3. Plot:** unit circle, $\circ$ for each zero, $\times$ for each pole, and the zeros at the origin if the numerator is the shorter polynomial.
- 4. Three anchor values,** which is all a not-to-scale sketch needs: $|H(1)|$ at $\omega = 0$, $|H(j)|$ at $\omega = \pi/2$, $|H(-1)|$ at $\omega = \pi$.
- 5. Peak at the pole angle, dip at the zero angle.** Join smoothly, then say in words whether it is low-pass, high-pass, band-pass or a notch.

The geometry, in one line

$$|H(e^{j\omega})| = |K| \frac{\prod_k |e^{j\omega} - z_k|}{\prod_k |e^{j\omega} - p_k|}$$

- Distances from the zeros multiply on top, distances from the poles divide underneath.
- A pole at radius r and angle θ makes $|H|$ peak at $\omega = \theta$, and the peak is sharper the closer r is to 1.
- A zero on the circle forces $|H|$ to **exactly zero** at its angle.

The checks that cost nothing

- 1.** $|H(1)|$ is the sum of the coefficients over the sum of the denominator coefficients; $|H(-1)|$ is the same with alternating signs. Two lines.
- 2.** Complex roots must be **conjugate pairs**. A lone complex pole is an arithmetic slip, every time.
- 3.** Product of the roots of $z^2 + \alpha z + \beta$ is β , sum is $-\alpha$. Check your factorisation against the coefficients before plotting.
- 4. If any pole has $|p| \geq 1$ and the system is stated causal, stop:** the ROC excludes the unit circle and there is no frequency response to draw.

1 Difference equation given: find $H(z)$, plot, sketch $|H|$

TOP 24/45 80 Ch 8, 72 Ash 7, 78 Bh 2+4, 81 Bh 3+7, 80 Bh 3+7, 81 Ba 3+7, 76 Ash 2+8, 79 Bh 3+7, 79 Ba 3+7, 74 Ash 3+7, 72 Ka 6, 72 Ch 2+4, 70 Ch 6, 71 Bh 6, 73 Ma 7, 69 Ch 2+2+6, 76 Bh 2+5, 75 Ch 2+7, 69 Bh 3+3+2+4, 70 Asa 2+6+2, 70 Ma 8

QUESTION AS PRINTED 2080 Chaitra · 2072 Ashwin · [8/7]

Plot Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.4y[n-1] + 0.25y[n-2] = x[n] + 0.5x[n-1]$ Both papers print exactly this, and **neither asks for a pole-zero plot**.

Five of the 24 ask for the magnitude response and nothing else: 80 Ch, 72 Ash, 72 Ka, 70 Ch and 71 Bh. The pole-zero plot is worth 2 or 3 marks in the others and **nothing in these**, so the plot still helps you draw the shape but does not earn on its own. Every other paper in this section asks for both.

Step 1: transform and divide

- Z-transform term by term, using $y[n-k] \leftrightarrow z^{-k}Y(z)$:

$$Y(z)(1 - 0.4z^{-1} + 0.25z^{-2}) = X(z)(1 + 0.5z^{-1})$$

$$H(z) = \frac{1 + 0.5z^{-1}}{1 - 0.4z^{-1} + 0.25z^{-2}} = \frac{z(z + 0.5)}{z^2 - 0.4z + 0.25}$$

- Multiplying by z^2/z^2 is what exposes the zero at the origin. **Do it before factoring, every time.**

Step 2: factor

- **Zeros**, from the numerator $z(z + 0.5)$:

$$z = 0 \quad \text{and} \quad z = -0.5$$

- **Poles**, from $z^2 - 0.4z + 0.25 = 0$:

$$z = \frac{0.4 \pm \sqrt{0.16 - 1}}{2} = 0.2 \pm j0.4583$$

- Check: product of the poles is $0.04 + 0.21 = 0.25$, the constant term. Sum is 0.4. Both agree.

- In polar form the pole pair is

$$|p| = \sqrt{0.2^2 + 0.4583^2} = \mathbf{0.5}, \quad \theta = \arctan \frac{0.4583}{0.2} = 66.4^\circ = 1.1$$

- $|p| = 0.5 < 1$, so the causal system is **stable** and the frequency response exists.
- $|p| = 0.5$ **exactly**, because the constant term of a monic quadratic is the product of the roots, and here that product is $|p|^2 = 0.25$.

Step 3: the three anchor values

- At $\omega = 0$, put $z = 1$: $|H| = \left| \frac{1 + 0.5}{1 - 0.4 + 0.25} \right| = \frac{1.5}{0.85} = \mathbf{1.765}$
- At $\omega = \pi/2$, put $z = j$: $|H| = \left| \frac{1 - 0.5j}{1 + 0.4j - 0.25} \right| = \frac{1.118}{0.850} = \mathbf{1.315}$
- At $\omega = \pi$, put $z = -1$: $|H| = \left| \frac{1 - 0.5}{1 + 0.4 + 0.25} \right| = \frac{0.5}{1.65} = \mathbf{0.303}$

Step 4: read the shape off the plot

- The pole pair sits at 66.4° , so $|H|$ **peaks** near there. The true peak is **1.96** at **50.9°** , pulled back towards $\omega = 0$ by the zero at $z = -0.5$ sitting on the far side.
- The zero at $z = -0.5$ lies at angle 180° , so $|H|$ is **smallest** at $\omega = \pi$, and 0.303 is the minimum.
- The zero at the origin is equidistant from every point of the circle and **does not shape the response at all**. It still belongs on the plot.
- Verdict: a **low-pass** response, rising a little to a broad hump around 50° before falling to 0.303 at $\omega = \pi$.

The same poles with the zero moved: 78 Bh and 81 Bh

Two more papers ask the identical question with $x[n] - 0.4x[n-1]$ on the right instead of $x[n] + 0.5x[n-1]$. **The denominator is unchanged, so the poles are the same $0.2 \pm j0.4583$** ; only the zero moves, from -0.5 to $+0.4$, that is from angle 180° to angle 0° .

- The zero now sits nearest $\omega = 0$, so it **pulls DC down** instead of $\omega = \pi$: $|H(1)| = 0.6/0.85 = \mathbf{0.706}$, $|H(j)| = \mathbf{1.267}$, $|H(-1)| = 1.4/1.65 = \mathbf{0.849}$.
- Peak **1.34** at **73.9°** , and the response is now **high-pass** in character.
- **One sign in the numerator flips the filter from low-pass to high-pass, with the poles untouched**. That is the lesson of the whole section.

Every other paper in this group

Same five steps. Poles from the y coefficients, zeros from the x coefficients, then the three anchor values.

Papers	Question as printed	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
80 Bh 3+7	Plot the pole-zero on the z-plane and draw Magnitude response (not to the scale) of an LTI system described by the equation, $y(n) = x(n) + 0.8x(n-1) + 0.8x(n-2) + 0.49y(n-2)$.	$-0.4 \pm j0.8$ (0.894)	± 0.7 (0.7)	5.10 0.55 1.96	low-pass, dip at 116°
79 Bh 3+7	Plot the pole-zero in z-plane and draw the magnitude response (not to the scale) of the equation of the system describe by difference equation: $y[n] - 0.35y[n-1] + 0.25y[n-2] = x[n] - 0.75x[n-1]$.	0.75, 0	$0.175 \pm j0.468$ (0.5)	0.28 1.51 1.09	high-pass, peak 1.53 at 83°
79 Ba 3+7	Plot the pole-zero in z-plane and draw magnitude response (not to the scale) of the system described by difference equation $y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] - 0.5x[n-1]$	0.5, 0	$0.15 \pm j0.421$ (0.447)	0.56 1.31 1.00	high-pass, peak 1.31 at 85°
74 Ash 3+7	Plot the pole-zero in z-plane and Draw Magnitude Response (not to the scale) of the system described by difference equation $y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.5x[n-1] + 0.6x[n-2] + 0.8x[n-3]$ third order in x , so three zeros, not two	$0.185 \pm j0.942$ (0.96), -0.869	$0.2 \pm j0.4$ (0.447)	3.63 0.56 0.19	low-pass, dip at 79°

Papers	Question as printed	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
72 Ka 6	Plot Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.5y[n-1] + 0.3y[n-2] = x[n] + 0.7x[n-1]$ no pole-zero plot asked: the response only	-0.7, 0	$0.25 \pm j0.487$ (0.548)	2.13 1.42 0.17	low-pass, peak 2.44 at 50°
72 Ch 2+4	Plot the pole-zero in z-plane and Draw Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.4y[n-1] + 0.1y[n-2] = x[n] + 0.6x[n-1]$	-0.6, 0	$0.2 \pm j0.245$ (0.316)	2.29 1.18 0.27	low-pass
70 Ch 6	Plot Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.3y[n-1] + 0.225y[n-2] = x[n] + 0.5x[n-1]$ no pole-zero plot asked; and note +0.5 where 71 Bh prints -0.4	-0.5, 0	$0.15 \pm j0.45$ (0.474)	1.62 1.35 0.33	low-pass, peak 1.80 at 55°
Papers	Question as printed	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
71 Bh 6	Plot Magnitude Response (not to the scale) of the system described by difference equation. $y[n] - 0.3y[n-1] + 0.225y[n-2] = x[n] - 0.4x[n-1]$ no pole-zero plot asked	0.4, 0	$0.15 \pm j0.45$ (0.474)	0.65 1.30 0.92	high-pass, peak 1.32 at 81°
73 Ma 7	Plot the pole-zero in z-plane and draw magnitude response (not to the scale) of the system described by difference equation $y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] + 0.7x[n-1]$	-0.7, 0	$0.15 \pm j0.421$ (0.447)	1.89 1.43 0.20	low-pass, peak 2.00 at 48°
Papers	Question as printed	Zeros	Poles ($ p $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Character
69 Ch 2+2+6	Why do we need Difference Equation? Draw Pole-zero in Z-Plane and plot magnitude response (not to the scale) of the system described by difference equation $y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.1x[n-1] - 0.06x[n-2]$	-0.3, 0.2	$0.2 \pm j0.4$ (0.447)	1.30 1.19 0.53	low-pass, peak 1.51 at 55°
76 Bh 2+5	Show the pole-zero in z-plane and plot magnitude response (not to the scale) of the system described by difference equation. $y[n] - 0.6y[n-1] + 0.35y[n-2] = x[n] - 0.1x[n-1] - 0.2x[n-2]$	0.5, -0.4	$0.3 \pm j0.510$ (0.592)	0.93 1.36 0.46	band-pass, peak 1.91 at 61°
75 Ch 2+7	Plot the pole-zero in z-plane and draw magnitude response (not to scale) of the system described by differential equation $y(n) - 0.3y(n-1) = 2x(n-2) + 0.7x(n-1) + 4x(n)$ "differential" for "difference", and the x terms are printed backwards , $n-2$ first	$-0.0875 \pm j0.702$ (0.707)	0.3 (0.3)	9.57 2.03 4.08	low-pass, dip at 99°
69 Bh 3+3+2+4	Explain how the poles and zeros of $H(z)$ affect the stability and the gain response of a system. Given $H(z)$ for a system with the difference equation $y[n] = x[n] + 1.21x[n-2] - 0.8y[n-1]$, plot its poles and zeros on the Z-plane, determine whether it is causal and stable and sketch its gain response.	$\pm j1.1$ (1.1)	-0.8 (0.8)	1.23 0.16 11.05	high-pass, deep dip at 90°
70 Asa 2+6+2 , 70 Ma 8	Given $H(z)$ for a system with the following difference equation: $y(n) = x(n) + x(n-2)$. Plot its poles and zeros in Z plane. Determine its magnitude response. Also, determine whether system is causal and stable. FIR, no feedback	$\pm j$ on the circle	none (all-zero)	2.00 0 2.00	$ H =$ $2 \cos \omega $, exact null

75 Ch prints $2x(n-2) + 0.7x(n-1) + 4x(n)$ **in that order** Read the delays, not the position on the page. The constant term is 4, so $H(z) = (4 + 0.7z^{-1} + 2z^{-2})/(1 - 0.3z^{-1})$ and $|H(1)| = 6.7/0.7 = 9.57$. Writing it as $2 + 0.7z^{-1} + 4z^{-2}$ instead gives 9.57 as well, because the coefficients still sum to 6.7, **but every other value is wrong** and the zeros move from $-0.0875 \pm j0.702$ to $-0.0875 \pm j0.351$.

70 Asa and 70 Ma also ask whether the system is causal and stable Yes to both, and the reason is one line: $y(n) = x(n) + x(n-2)$ uses only present and past inputs, so it is causal; its impulse response is $h = \{1, 0, 1\}$, a finite sum, so $\sum |h[n]| = 2 < \infty$ and it is stable. **Every FIR system is stable**, which is why the question is worth only 2 marks.

81 Ba and 76 Ash: the pole is at $z = 2.75$ $y(n) = 0.67x(n) - 0.3x(n-1) + 2.75y(n-1)$ rearranges to $y(n) - 2.75y(n-1) = 0.67x(n) - 0.3x(n-1)$, so $H(z) = (0.67z - 0.3)/(z - 2.75)$: one zero at $z = 0.448$ and **one pole at $z = 2.75$, far outside the unit circle**. If the system is causal its ROC is $|z| > 2.75$, which does not contain the unit circle, so **the frequency response does not exist** and neither the magnitude (81 Ba) nor the phase (76 Ash) can be drawn. Only the anticausal reading, ROC $|z| < 2.75$, is stable; on that reading $|H|$ climbs gently from 0.211 at $\omega = 0$ to 0.259 at $\omega = \pi$. **State which reading you are using**. A pole at 0.275 rather than 2.75 would make the causal system stable, and is the likeliest intended value.

2 Poles and zeros given directly

TOP 11/45 82 Ba **3+7**, 80 Ba **4+6**, 82 Bh **2+8**, 76 Ch **2+6**, 75 Ash **2+8**, 81 Ch **8**, 73 Bh **8**, 74 Ch **2+8**, 75 Bh **2+6**, 73 Ch **2+8**, 71 Ch **2+8**
no algebra at all: the factoring is done for you

QUESTION AS PRINTED 2080 Baishakh · 2082 Bhadra · [4+6]

A filter has poles at $0.45 \pm j1.6$ and zeros at $0.58 \pm j2.06$. Map them in the z -plane and plot the magnitude response (not to scale).

Step 1: radii and angles, which is all the geometry needs

- **Poles** $0.45 \pm j1.6$:

$$|p| = \sqrt{0.45^2 + 1.6^2} = \mathbf{1.662}, \quad \theta_p = \arctan \frac{1.6}{0.45} = 74.3^\circ$$

- **Zeros** $0.58 \pm j2.06$:

$$|z| = \sqrt{0.58^2 + 2.06^2} = \mathbf{2.140}, \quad \theta_z = \arctan \frac{2.06}{0.58} = 74.3^\circ$$

- **Poles and zeros share the same angle**, 74.3° , and the zeros sit further out. So they partly cancel: the peak the poles want to make is flattened by the dip the zeros want to make.
- Both lie **outside** the unit circle, so nothing is close to the circle and **the response is gentle, with no sharp feature anywhere**.
- That is the answer the 6 marks are for. The sketch is nearly flat.

Step 2: the three anchor values

- Build $H(z) = \frac{(z - 0.58 - j2.06)(z - 0.58 + j2.06)}{(z - 0.45 - j1.6)(z - 0.45 + j1.6)} = \frac{z^2 - 1.16z + 4.578}{z^2 - 0.9z + 2.763}$, using (sum, product) of each conjugate pair.
- $|H(1)| = \left| \frac{1 - 1.16 + 4.578}{1 - 0.9 + 2.763} \right| = \frac{4.418}{2.863} = \mathbf{1.544}$
- $|H(j)| = \mathbf{1.902}$, $|H(-1)| = \left| \frac{1 + 1.16 + 4.578}{1 + 0.9 + 2.763} \right| = \frac{6.738}{4.663} = \mathbf{1.446}$
- Peak **2.04** at 74° , exactly the pole angle. Range 1.45 to 2.04: **a shallow bump, not a resonance**.

Step 3: the phase, when asked

82 Ba asks for the phase as well. Add the angle of each vector from a zero and subtract the angle of each vector from a pole: $\arg H = \sum_k \arg(e^{j\omega} - z_k) - \sum_k \arg(e^{j\omega} - p_k)$. With conjugate pairs the imaginary parts cancel at $\omega = 0$ and $\omega = \pi$, so **the phase is zero at both ends** and odd in between.

The rest of the family

Papers	Question as printed	Poles ($ p $)	Zeros ($ z $)	$ H $ at $0, \frac{\pi}{2}, \pi$	Shape
82 Ba 3+7 , 80 Ba 4+6 , 82 Bh 2+8	Draw the poles and zeros in the z-plane for the system with poles at $0.45 \pm j1.6$ and zeros at $0.58 \pm j2.06$. Also plot the magnitude response (not in scale) of the system. 80 Ba prefixes "Differentiate between FIR system and IIR System"; 82 Ba asks for the phase response as well	$0.45 \pm j1.6$ (1.662)	$0.58 \pm j2.06$ (2.140)	1.54 1.90 1.45	shallow bump, peak 2.04 at 74°
76 Ch 2+6 , 75 Ash 2+8 , 81 Ch 8 , 73 Bh	Draw the pole-zero in the z-plane for a system with poles at $0.45 \pm j1.06$ and zeroes at $0.58 \pm j2.06$. Also plot the magnitude response (not to the scale) of the system. $j1.06$, not $j1.6$: one digit pulls the poles in and the peak goes from 2 to 11.5. 75 Ash omits "not to scale"	$0.45 \pm j1.06$ (1.152)	$0.58 \pm j2.06$ (2.140)	3.10 3.93 2.09	real peak 11.50 at 67°
74 Ch 2+8	The poles of a system are located at: $0.45 - 0.77i$ and $-2 \pm 0.3i$. Map the poles and zero in the z-plane and plot the magnitude response of the system. a lone complex pole , not a pair, and no zeros are given at all	$0.45 - j0.77$ (0.892), $-2 \pm j0.3$ (2.022)	none	0.20 0.18 0.63	rises towards $\omega = \pi$
75 Bh, 71 Ch 2+8	The poles of a system are located at: $0.45 + 0.77i$ and $-2 \pm 0.3i$ and zeroes at: $1.2 \pm 3i$. Map the poles and zero in the z-plane and plot the magnitude response of the system.	$0.45 \mp j0.77$ (0.892), $-2 \pm j0.3$ (2.022)	$1.2 \pm j3$ (3.231)	1.81 1.76 8.76	high-pass, peak 8.76 at π
73 Ch 2+8	Define the difference equation with example. The Poles of a system are located at: $0.45 + 0.77i$ and $2 \pm 0.7i$ and zeros at: $1.2 \pm 0.43i$. Plot the magnitude response of this system. +2, not -2 , which is what moves its bump	$0.45 + j0.77$ (0.892), $2 \pm j0.7$ (2.119)	$1.2 \pm j0.43$ (1.275)	0.27 0.43 0.37	broad bump at 72°

Every pole in this section except $0.45 \pm j0.77$ lies outside the unit circle $|p|$ runs 1.152, 1.662, 2.022, 2.119. So **none of these systems is both causal and stable**. The questions only ask you to map the points and sketch $|H|$, which is legitimate as long as the ROC is taken to be an annulus that contains the unit circle, making the system stable but **not causal**. **Say so in one line and the mark is safe**. The $0.45 \pm j0.77$ pair, at $|p| = 0.892$, is the only one inside.

74 Ch, 75 Bh, 73 Ch and 71 Ch each print a lone complex pole They give $0.45 - 0.77i$ or $0.45 + 0.77i$ with **no conjugate**, alongside a pair $-2 \pm 0.3i$ or $2 \pm 0.7i$ that is properly conjugate. **A difference equation with real coefficients cannot have an unpaired complex pole**, so the intended pole is the pair $0.45 \pm j0.77$ and the printed sign is a dropped $\pm$. Restoring the conjugate changes the numbers, so **state which reading you used**: with the pair restored, 74 Ch has $|H| = 0.12, 0.21, 0.34$ at $0, \pi/2, \pi$ and peaks at 0.80 near 60° . The values tabulated above are for the question exactly as printed.

Where $0.45 \pm j0.77$ comes from The local worked example these questions are built on specifies its poles and zeros as a *radius and an angle*: a zero of radius 0.9 at 1.0376 rad is $0.9 \cos(1.0376) + j0.9 \sin(1.0376) = 0.4575 + j0.7751$, **which the source rounds to $0.45 + j0.77$** , and a pole of radius 0.892 at 2.5158 rad is $-0.728 + j0.522$. **Recognising the pair saves a conversion**, and it explains why the same two decimals keep reappearing across six different papers.

3 All-pole systems, and the steady-state response to a sinusoid

PIN 3/45 69 Bh **6**, 67 Mng **10**, 68 Bh **7**

QUESTION AS PRINTED 2068 Bhadra · [7]

For the system $y[n] - \frac{10}{24}y[n-1] + \frac{1}{24}y[n-2] = x[n]$, find the frequency response, then determine $y[n]$ for the input $x[n] = \sin(\frac{\pi}{3}n) + \sin(\frac{\pi}{5}n)$.

Step 1: the frequency response

- There is no x delay, so the numerator is 1: the system is **all-pole**.

$$H(z) = \frac{1}{1 - \frac{10}{24}z^{-1} + \frac{1}{24}z^{-2}} = \frac{z^2}{z^2 - \frac{5}{12}z + \frac{1}{24}}$$

- Factor: $z^2 - \frac{5}{12}z + \frac{1}{24} = (z - \frac{1}{4})(z - \frac{1}{6})$, so the **poles are $\frac{1}{4}$ and $\frac{1}{6}$** , both real, both well inside the circle. Two zeros sit at the origin.

$$H(e^{j\omega}) = \frac{1}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{1}{6}e^{-j\omega})}$$

- $|H|$ falls from 1.60 at $\omega = 0$ to 0.686 at $\omega = \pi$: a gentle **low-pass**, with no peak because both poles are far from the circle.

Step 2: use the eigenfunction result, do not convolve

- A sinusoid in gives the same sinusoid out, scaled by $|H|$ and shifted by $\angle H$ at that frequency. Evaluate H twice and stop.
- At $\omega = \pi/3$: $|H| = \mathbf{1.1955}$, $\angle H = \mathbf{-0.3987}$ rad = -22.85° .
- At $\omega = \pi/5$: $|H| = \mathbf{1.4159}$, $\angle H = \mathbf{-0.2949}$ rad = -16.90° .

$$y[n] = 1.1955 \sin\left(\frac{\pi}{3}n - 0.3987\right) + 1.4159 \sin\left(\frac{\pi}{5}n - 0.2949\right)$$

- **The lower frequency is amplified more**, which is the low-pass character showing up in the answer.

69 Bh: the same method, one line shorter

- $y[n] - 0.8y[n-1] + 0.15y[n-2] = x[n]$ gives $z^2 - 0.8z + 0.15 = (z - 0.5)(z - 0.3)$, so the **poles are 0.5 and 0.3** with a double zero at the origin.
- $|H|$: **2.857** at $\omega = 0$, 0.857 at $\pi/2$, **0.513** at π . Monotonically falling, so **low-pass**, peak at DC.

67 Mng as printed has a pole exactly on the unit circle The paper writes $y[n] - \frac{5}{6}y[n-1] - \frac{1}{6}y[n-2] - x[n] = 0$. That denominator factors as $z^2 - \frac{5}{6}z - \frac{1}{6} = (z - 1)(z + \frac{1}{6})$, putting **a pole at $z = 1$** . A pole on the circle means $|H(e^{j0})|$ is infinite, the system is only marginally stable and it is not BIBO stable, so strictly **there is no frequency response to plot**. Change the one sign, $-\frac{1}{6} \rightarrow +\frac{1}{6}$, and the poles become $\frac{1}{2}$ and $\frac{1}{3}$, giving an ordinary low-pass with $|H(1)| = 1/[(1 - \frac{1}{2})(1 - \frac{1}{3})] = 3$. **68 Bh asks the identical question with the + sign** and poles $\frac{1}{4}, \frac{1}{6}$, which is strong evidence that the minus is a misprint. Answer the + version and note the printed one in a line.

4 Zero-input, zero-state and total response

PIN 2/45 78 Bh **4**, 66 Ma **10**

QUESTION AS PRINTED 2066 Magh · [10]

A system is characterized by $y(n) - 0.5y(n-1) - 0.25y(n-2) - x(n) = 0$, the input is $x(n) = 0.2^n u(n)$ and the initial conditions are $y(-1) = 2$, $y(-2) = 4$. Find (a) the zero input response, (b) the zero state response, (c) the total response, (d) the system function $H(z)$, (e) the poles of $H(z)$.

(d) and (e) first, because everything else uses them

$$H(z) = \frac{1}{1 - 0.5z^{-1} - 0.25z^{-2}} = \frac{z^2}{z^2 - 0.5z - 0.25}$$

- Roots: $z = \frac{0.5 \pm \sqrt{0.25 + 1}}{2} = \frac{1 \pm \sqrt{5}}{4}$, that is

$$p_1 = \mathbf{0.8090}, \quad p_2 = \mathbf{-0.3090}$$

- Both have $|p| < 1$, so the causal system is **stable**.

(a) Zero-input response: the input is switched off, the initial conditions drive it

- The form is fixed by the poles: $y_{zi}[n] = C_1(0.8090)^n + C_2(-0.3090)^n$.
- Impose $y(-1) = 2$ and $y(-2) = 4$ on that form and solve the pair:

$$C_1 = 1 + \frac{2}{\sqrt{5}} = \mathbf{1.8944}, \quad C_2 = 1 - \frac{2}{\sqrt{5}} = \mathbf{0.1056}$$

$$y_{zi}[n] = 1.8944(0.8090)^n + 0.1056(-0.3090)^n, \quad n \geq 0$$

- Quick check by running the recurrence $y[n] = 0.5y[n-1] + 0.25y[n-2]$ with no input: $y[0] = 0.5(2) + 0.25(4) = 2$, $y[1] = 1.5$, $y[2] = 1.25$, $y[3] = 1$. The closed form gives the same four numbers.

(b) Zero-state response: the initial conditions are zero, the input drives it

- $X(z) = \frac{1}{1 - 0.2z^{-1}}$, so $Y_{zs}(z) = H(z)X(z)$ has three poles: 0.8090, -0.3090 and 0.2.

- Partial-fraction $Y_{zs}(z)/z$ and multiply back:

$$y_{zs}[n] = 0.9612 (0.8090)^n + 0.1678 (-0.3090)^n - 0.1290 (0.2)^n, \quad n \geq 0$$

- The three residues sum to 1, which must happen because $y_{zs}[0] = 1$. **That is the free check.** The last residue is exactly $-\frac{4}{31}$.

(c) Total response

$$y[n] = y_{zi}[n] + y_{zs}[n] = 2.8557 (0.8090)^n + 0.2734 (-0.3090)^n - 0.1290 (0.2)^n$$

- First few values: 3, 2.2, 1.89, 1.503, 1.2256. Running the original recurrence with $y(-1) = 2$, $y(-2) = 4$ and $x(n) = 0.2^n$ gives the same sequence.
- **Natural against forced is a different split from zero-input against zero-state.** The natural part is everything in $(0.8090)^n$ and $(-0.3090)^n$, which draws from both; the forced part is only the $(0.2)^n$ term.

78 Bh gives no initial conditions $y[n] - 3y[n-1] - 4y[n-2] = x[n]$ has $z^2 - 3z - 4 = (z-4)(z+1)$, so the zero-input response is $y_{zi}[n] = C_1(4)^n + C_2(-1)^n$ and **that form is the whole answer**, since no $y(-1)$ or $y(-2)$ is printed to pin C_1 and C_2 down. Worth adding: $|4| > 1$ and $|-1| = 1$, so the zero-input response **never decays** and the system is unstable. Say it, because the 4 marks are for recognising the form and its behaviour.

5 Linear phase

PIN 3/45 70 Asa **2+4**, 67 Mng **6**, 69 Bh **6** · the derivation for general M is in §3.6

QUESTION AS PRINTED 2070 Ashad · [2+4]

What is the condition satisfied by a linear phase FIR filter? Show that the filter with $h(n) = \{-1, 0, 1\}$ is a linear phase filter.

The condition, for the 2 marks

$$h[n] = \pm h[M-1-n], \quad 0 \leq n \leq M-1$$

- The impulse response must be **symmetric (+)** or **antisymmetric (-)** about its midpoint $\alpha = \frac{M-1}{2}$.
- That α is then the constant group delay in samples.

Now show the phase really is linear

Check the given $h(n)$

- $M = 3$, so $M-1-n$ maps $0 \leftrightarrow 2$ and fixes 1.
- $h[0] = -1$ and $h[2] = +1$, so $h[0] = -h[2]$; and $h[1] = 0 = -h[1]$.
- So $h[n] = -h[2-n]$: the sequence is **antisymmetric**, and the condition is met with the minus sign.

$$H(e^{j\omega}) = \sum_{n=0}^2 h[n]e^{-j\omega n} = -1 + e^{-2j\omega}$$

- Factor out the midpoint delay $e^{-j\omega}$, which is the standard trick:

$$H(e^{j\omega}) = e^{-j\omega}(e^{-j\omega} - e^{+j\omega}) = e^{-j\omega}(-2j \sin \omega) = 2 \sin \omega e^{-j(\omega+\pi/2)}$$

$$|H(e^{j\omega})| = 2|\sin \omega|, \quad \arg H(e^{j\omega}) = -\omega - \frac{\pi}{2}$$

- The phase is a **straight line** in ω with slope -1 , so

$$\text{grd } H = -\frac{d}{d\omega} \left(-\omega - \frac{\pi}{2} \right) = 1 \text{ sample} = \alpha = \frac{M-1}{2} = 1 \checkmark$$

- **The constant $-\pi/2$ is why this is generalized linear phase.** It comes from the j in $-2j \sin \omega$, that is from the antisymmetry, and it does not affect the group delay, so it causes no distortion.
- The magnitude $2|\sin \omega|$ is zero at $\omega = 0$: an antisymmetric filter always kills DC, which is why this one is a differencer.

67 Mng: show that $h[n] = h[N - 1 - n]$, $0 \leq n \leq N - 1$, is linear phase This is the general case of the same argument and it is worked in full in §3.6: pair each term with its mirror, factor out $e^{-j\omega\alpha}$ with $\alpha = (N - 1)/2$, and what is left is a real cosine sum. **Real bracket, so the whole phase is the exponential**, giving $\arg H = -\alpha\omega$. 69 Bh asks for the same derivation restricted to odd M , which is the easier case because the midpoint sample $h[\alpha]$ stands alone instead of pairing.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. $\rightarrow$ reads “leads to / therefore”, $+$ reads “and”, $=$ reads “is”. $\checkmark$ means the property holds, $\times$ that it does not.

Short	Means	Short	Means
DSP	digital signal processing	DSAP	digital signal analysis and processing
DT	discrete-time	CT	continuous-time
LTI	linear time-invariant	LSI	linear shift-invariant, the same thing
TI	time-invariant	BIBO	bounded input, bounded output
LCCDE	linear constant-coefficient difference equation	ROC	region of convergence
DTFS	discrete-time Fourier series	DTFT	discrete-time Fourier transform
CTFT	continuous-time Fourier transform	DFS	discrete Fourier series
DFT	discrete Fourier transform	IDFT	inverse DFT
FFT	fast Fourier transform	DIT	decimation in time
DIF	decimation in frequency	MAC	multiply and accumulate
FIR	finite impulse response	IIR	infinite impulse response
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
BLT	bilinear transformation	II	impulse invariance
A/D	analog to digital	D/A	digital to analog
S/H	sample and hold	ZOH	zero-order hold
SNR	signal to noise ratio	PYQ	past-year question
coeff	coefficient	freq	frequency
eqn	equation	no.	number
ops	operations	attr	attribute
w/	with	w/o	without
Eg	for example	Vs	compare / differentiate
+Fig	a figure is wanted	(I)	importance / need
(+)	advantage	(-)	disadvantage

Exam paper codes

Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months:** 2075 Ashwin and 2070 Ashad are separate papers.

How a year code is written

- **Bold** = Regular exam. Plain = Back exam.
 - “New Back (2066 & Later Batch)” counts as Back.
 - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 Bh is EX 753 Regular, 74 Ma is EX 753 Back.
- **69 Bh and 69 Bh are two different papers**, one in each course.

Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.